



No back button on Android Q

Android Q to ditch back button and will probably replace it with iOS-like gestures for navigation. <https://digit.in/mar19-85>



Lossless 10x zooming from Oppo

Oppo to unveil 10x Lossless Zoom camera tech, could be showcased on the Oppo F19 and F19 Pro smartphones. <https://digit.in/mar19-86>

MSI GE75 Raider 8SG

Now with real-time ray tracing

While every manufacturer unveiled an NVIDIA RTX GPU laptop at CES 2019, MSI has been the first to bring it to India. The MSI GE75 Raider 8SG retains the same Intel Core i7-8750H CPU while upgrading to the newer NVIDIA RTX 2080 GPU. Most of the other components, which includes the RAM, hard drives, SSDs, etc., have remained the same with the exception of the cooling system which obviously had to be upgraded to keep the ridiculous RTX GPU cool.

MAX-P vs MAX-Q

The RTX 2080 comes in two variants, the MAX-P and the MAX-Q. The MAX-P is considered the regular RTX 2080 for laptops while the MAX-Q GPU is clocked much lower to make it viable for slim form factor machines. The difference between the two is massive. The desktop RTX 2080 is clocked at 1515 MHz and has 2944 CUDA cores whereas the MAX-P variant is clocked at 1380 MHz and lastly, the MAX-Q is clocked at 735 MHz. The core configurations i.e. shaders, ROPs and Texture Mapping Units remain the same. The TDP for the two configurations are also vastly different, 80W and 150W. Let's not even get into power consumption figures. Simply put, go for the normal variant or the MAX-P for better performance.

Build Quality

The entire top side of the laptop is aluminium and the same can be said for the keyboard chassis. The other parts are plastic. The lid has a smooth matte black texture and makes fingerprints stand out as clear as day.

MSI is using WD Black and Seagate Barracuda Pro drives in the GE75 Raider 8SG. The unit we received had



Price
₹2,49,990



PERFORMANCE.....85
FEATURES.....81
BUILD78

the latter. You can also find the two Kingston A1000 480 GB SSDs that are in RAID 0 configuration towards the right edge of the motherboard. The sides are chock-full on I/O ports and they've left no space unused. This would be one

of the few laptops we've seen that's still sporting an SD card slot. We're not pointing it out as a bad thing, in fact, we'd take as many kinds on I/O options as possible on a laptop.

The key thing to focus on here is the cooling system. The CPU can be seen on the right hand side with two heatpipes running over it while a thicker third one is used for the VRM before slithering onto the GPU. The GPU, on the other hand, has six heatpipes running over a heat spreader that covers both the GPU as well as the VRM for the GPU. We can see a similar design on the previous GE75 and GE73 Raider but with one less heatpipe. So the current gen GE75 does have improved cooling but what stands to be seen is if it's sufficient in Indian conditions given that the GPU is practically a hot plate.

Performance

The MSI GE75 8SG uses two Kingston A1000 SSDs, that are based on the Phison E8 controller, in RAID 0 configuration. In synthetic benchmarks,

the RAID 0 configuration tops out at 3 GBps and the read speeds hits 1.5 GBps. The read speeds are considerably lower than the GT75 which averages around 2.4 GBps using two SM961 in RAID 0.

Gaming

Time Spy scored 8792 points while, TimeSpy Extreme clocked in at 3994. Fire Strike - 19144, Fire Strike Extreme - 11034, Sky Diver - 40180 and in VR Benchmark Blue Room - 3044.

Coming to the gaming benchmarks, we managed to get 67 FPS in Assassin's Creed Odyssey, 95 FPS in Shadow of the Tomb Raider, about 172 FPS in PUBG, and 73 FPS in Deus Ex Mankind Divided. All of these games were run on Ultra settings at 1080p resolution. Overall, there's about a 11-13% increase in gaming FPS on RTX 2080 MAX-P as opposed to the GTX 1080 laptop GPU. This is more or less the difference between the desktop reference cards for the same GPUs as well. The added benefit here being that you get to experience real-time ray-tracing in video games. The downside? There really aren't that many games which have implemented these new features.

Verdict

The MSI GE75 Raider 8SG is a good laptop for folks who wish to try out NVIDIA's new RTX GPUs. However, the cooling solution seems a bit underpowered for a hot country such as India. Overall, we feel that the MSI GE75 Raider 8SG is a great upgrade.

—Mithun Mohandas

SPECIFICATIONS

PROCESSOR: Intel Core i7-8750H | PLATFORM: HM370 | RAM: 16 GB 2666 MHz | SCREEN SIZE: 17.3-inches 120Hz | SCREEN RESOLUTION: 1920x1080 | GPU: GTX 2080 8 GB GDDR6 | SSD: 2x Kingston A1000 480 GB RAID 0 | HDD: Seagate 1TB 7200 RPM | BATTERY: 6-cell 65 Whr | AC ADAPTER: 280 W | WEIGHT: 2.66 Kg

CONTACT

ACRO ENGINEERING COMPANY | PHONE: 011-40525645 | EMAIL: saini@smcinternational.in | WEBSITE: <http://www.smcinternational.in>